

SUSTAINABLE DEVELOPMENT GOALS PLAN

Team The Frustrated Team

Members Benjamin | Anastasia | Anna | Temidayo | Ignacio | Traian

1. Category A: Sustainable Research and Innovation

SDG 9: Industry, Innovation and Infrastructure

Overall contribution. The platform strengthens materials research by making complex many-body systems easier to model, compare and reproduce. It can simulate Hamiltonians and wavefunctions efficiently, as opposed to classical computers, thus reducing unnecessary experimental iterations and connecting theoretical, computational and experimental teams.

Target 9.4: Sustainable industrial upgrading.

Support more sustainable industrial and research infrastructure. Simulation can reduce unsuccessful synthesis, testing, and characterisation cycles.

Target 9.5 Enhance scientific research.

Enhance scientific research and technological capability. Reusable models and benchmarking workflows allow more researchers to investigate advanced quantum materials.

Target 9.b Support domestic technology development.

Support domestic technology development. Cloud-based access can enable institutions without dedicated quantum hardware or neutron-scattering facilities to participate.

Expected impact: Faster hypothesis testing, focused experiments, lower barriers to research.

Priority indicators

- Physical synthesis and testing cycles avoided
- Percentage of candidate space screened virtually

SDG 12: Responsible Consumption and Production

Overall contribution. Simulation filtering can help materials research move from trial and error development toward more precise candidate selection.

Target 12.6 Encourage companies to adopt sustainable practices.

A future commercial platform could report estimated material, lab energy and facility use savings.

Target 12.a Strengthen scientific and technological capacity.

Shared simulation tools can help research teams examine lower-impact, more abundant, or more recyclable alternatives before fabrication.

Expected impact: Less sample waste, reduced use of cryogenic equipment, earlier identification of tolerant and resource-efficient candidates.

Priority indicators

- Material usage
- Laboratory/cryogenic hours avoided

Real-world example

Large logistics companies can use connected sensors and asset-tracking platforms to monitor the location and condition of goods, vehicles, or equipment. Amazon Web Services, for example, provides IoT and location tools for centralised asset tracking.

For our project, the relevant opportunity is the development of sensitive magnetic or condition-monitoring components that could **improve equipment visibility, reduce misplaced assets, support predictive maintenance, and limit unnecessary journeys or replacement.**

Commercial pathway

1 Research demo Validated simulations for material observables, supporting experimental planning.	2 Screening tool Parameter sweeps, candidate comparison, error estimates, sustainability metrics.	3 Industry pilot Adapted workflow to focused uses: sensing, low-energy logic, thermal management.
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2. Category B: Energy and Climate Applications

SDG 7: Affordable and Clean Energy

Overall contribution. The platform can accelerate investigation of magnetic materials whose collective behaviour may support lower-energy sensing, information processing and cooling.

Target 7.3 Improve energy efficiency..
Relevant research directions include low-loss magnetic components, spintronic memory, magnonic communication, magnetic refrigeration, sensing for energy infrastructure.

Expected impact: Lower switching energy, reduced electrical losses, improved equipment monitoring, efficient low-temperature thermal management.

Priority indicators

- Predicted energy reduction against classical methods
- Simulation energy used per candidate

Real-world example

Modern electricity networks use sensors, communication systems, and analytics to observe the condition of grid assets and respond to faults or disturbances.

A sufficiently sensitive magnetic sensor could contribute to **monitoring current flow, transformers, motors, cables, or power electronics**. Earlier fault detection can reduce downtime, prevent damage, and improve the use of existing infrastructure.

SDG 13: Climate Action

Overall contribution. The project supports climate action by improving the search for efficient electronics, sensing systems, thermal technologies and materials with lower critical mineral requirements.

Target 13.2 Integrate climate change measures into national policies.
The project contributes by screening magnetic behaviours relevant to lower-loss electronics, grid sensing, predictive maintenance and thermal management. These applications could support lower operational emissions and more resilient energy infrastructure.

Expected impact: Better sensing can extend equipment life, reduce leakage and avoid premature replacement. Efficient computing and thermal technologies would reduce operational emissions.

Priority indicators

- Estimated lifecycle CO₂e benefit of a pilot
- Percentage of simulation energy from low-carbon sources

Real-world example

State Grid Corporation of China uses sensors, automated inspection, and analytics to monitor grid equipment and improve fault diagnosis.

Our project supports this direction by screening magnetic responses for future current, transformer, cable, and power-electronics sensors, helping reduce downtime and unnecessary maintenance.

3. Category C: Education and Partnerships

SDG 4: Quality Education

Overall contribution. The project turns advanced quantum-material concepts into practical learning, connecting physics, code and experiment, for a modernization of the scholar curricula.

Target 4.4 Increasing technical skills among youth.

The project provides hands-on training in quantum simulation, CMP, scientific programming, experiment and sustainability assessment through student projects and cloud-compatible scientific demonstrations.

Expected impact: Improved quantum literacy, interdisciplinary learning, and broader access to citizen science research without requiring specialist hardware.

Priority indicators

- Learners completing the simulation module
- Institutions using the educational package
- Learning improvement and student satisfaction

Real-world example

IBM Quantum Learning shows how cloud access and guided courses can make quantum technologies accessible without specialist hardware. Our project applies the same model to quantum materials through **reproducible notebooks linking physics, simulation, observables and sustainability**.

SDG 17: Partnerships for the Goals

Overall contribution. Scientific validation and commercial translation require collaboration between quantum-computing teams, condensed-matter physicists, experimentalists, materials researchers, educators, cloud providers, and prospective technology users.

Target 17.6 Inter-institutional cooperation.

The project supports this target through shared quantum-material models, open datasets, reproducible simulations, and joint validation between quantum-computing teams, materials researchers, labs, educators and industrial partners.

Expected impact: Less duplicated work, faster validation, wider access to infrastructure, and a clearer route from open science to responsible commercial pilots.

Priority indicators

- Active academic, industrial, or international partners
- Shared datasets, models, or benchmark releases

Real-world example

The European Quantum Technologies Flagship connects research, industry, education, and public funding to accelerate quantum innovation. Our project follows this model by **linking simulation teams, experimental laboratories, computing partners and application users**.